# STAT 330-001

40142034 22

## Seahorn, Dallin

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Test: 638395 Form: A
Entry: 05:08pm 10/20/2026
Employee: Justin E.

Wadsworth, Spencer Gordon
Checked out:

**SPECIAL INSTRUCTIONS**

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

**PLEASE TURN OFF YOUR CELL PHONE**

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# STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

**Instructions**

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

## PART I

**Question 1:** **(Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

**Question 2:** **(TRUE/FALSE)** If the relationship between $x$ and $y$ is nonlinear, a simple linear regression model may still be feasible after transforming the data.

• TRUE

• FALSE

**Question 3:** **(Circle one answer)** When adding an additional variable to a linear regression model, the model's $R^2$ ... Not adjusted

(A) always increases or remains unchanged. ✓

(B) always decreases or remains unchanged. ✗

(C) may increase or decrease depending of the additional variable. ✗

**Question 4:** **(Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated. ✓

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients. ✓

(C) It can result in larger standard errors, and thus greater uncertainty in estimates. ✓

(D) It can cause the coefficient of determination ($R^2$) to become negative. ✗

**Question 5: (Circle all that apply)** Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection. ✗

(B) Cook's distance can be used to identify influential observations. ✓

(C) Transformations of the $x$'s or $y$ may help alleviate problems with linearity, equal variance, or normality. ✓

(D) The least-squares estimate of $\beta$ can be obtained by fitting a simple linear regression for each $x_k$ only. ✗

(E) The assumptions on the error variables (i.e., the $\epsilon_i$) in a multiple linear regression model are exactly the same as they are for simple linear regression. ✓

**Question 6: (Circle one answer)** What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data. ✗

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model. ✓

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process. ✗

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models. ✗

**Question 7: (Circle all that apply)** In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection ✓

(B) Their coefficient estimates are less biased than those of OLS ✗ introduce bias

(C) They produce more accurate confidence intervals and hypothesis tests ✗

(D) Ridge regression often handles multicollinearity well ✓

(E) They can handle datasets where the number of predictors exceeds the number of observations ✓

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions ✓

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

They shrink variance and introduce Bias to find the smallest values for the residuals. OLS is when lambda = 0. Ridge takes lambda and shrinks values close to zero, but not exactly 0. Lasso can shrink values all the way to 0, causing some predictors to not be included in a given model.

Ridge: $SSE + \lambda \sum B_j^2$

Lasso: $SSE + \lambda \sum |B_j|$

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$\frac{(2.4)^2}{3+1}\left(\frac{.25}{1-.25}\right) = 0.48$

$\frac{4}{n} = \frac{4}{100} = .04$

**(Circle one answer)** Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated. ✓

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model. X

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50. X

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated. X

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(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50. X

(G) Cook's distance cannot be calculated without knowing the observation's original residual. X

**Question 10:**

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = (X^T X)^{-1} X^T Y$

(B) What is the conceptual difference between $\widehat{\beta}$ and $\beta$?

In matrix notation, $\beta$ is place holders for the true $\beta$ population parameters, $\hat{\beta}$ is our estimated coefficients that we get from using the above formula.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

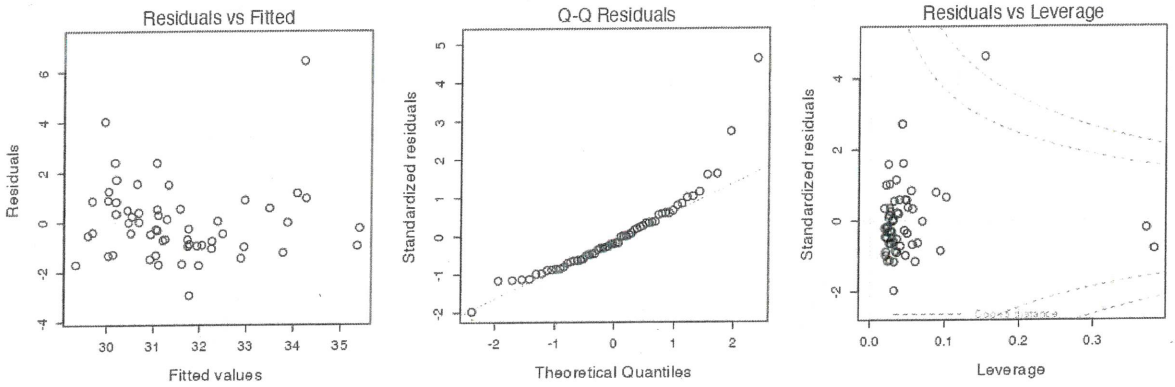
Least squares is used to select the coefficient vector that minimizes SSE. $\hat{Y} = X\hat{\beta}$ and SSE is the summation of our observed residuals minus our estimated residuals. We select the vector that minimizes this residual summation.

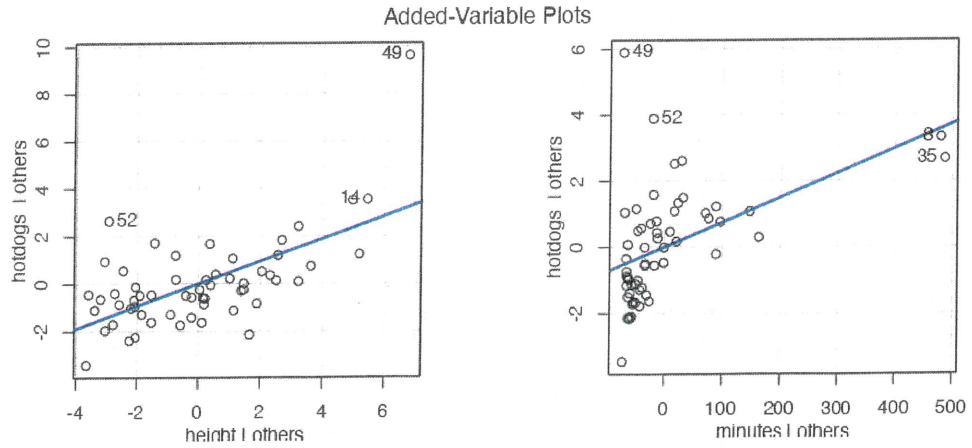
# PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

$\hat{\beta}_1 = .472892$ Holding $\beta_2$ (minutes trained) constant, we expect with an increase in height of an inch, that a contestant will eat .472892 more hotdogs.

**Question 12:** **(Circle all that apply)** Which of the following are true?

$CI = \beta_2 \pm t^* SE$

$.007417 \pm t^* .001871$

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0 ✗

(B) The 95% confidence interval of $\beta_2$ does contain 0 ✓

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$ ✗ $\hat{\beta}_1 = .4279$ not true $\beta_1$

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Looking at the Residuals vs Fitted plot, it looks like there is some curvature implying nonlinearity

(B) There is independence between each observation

Typically one person's consummation of hotdogs is independent from another, so independace is met.

(C) The errors are normally distributed

The Shapiro W.ilk test shows a low pvalue indicating non-normality. However, we must also look at the Q-Q plot which shows the tails being skewed, implying possible non-normality.

(D) The errors have constant variance (homoscedasticity)

According to the BP test, there doesn't seem to be unequal variance. With a quick look at Residuals Fitted plot, we can see that there seem to be equal variance with a few outliers

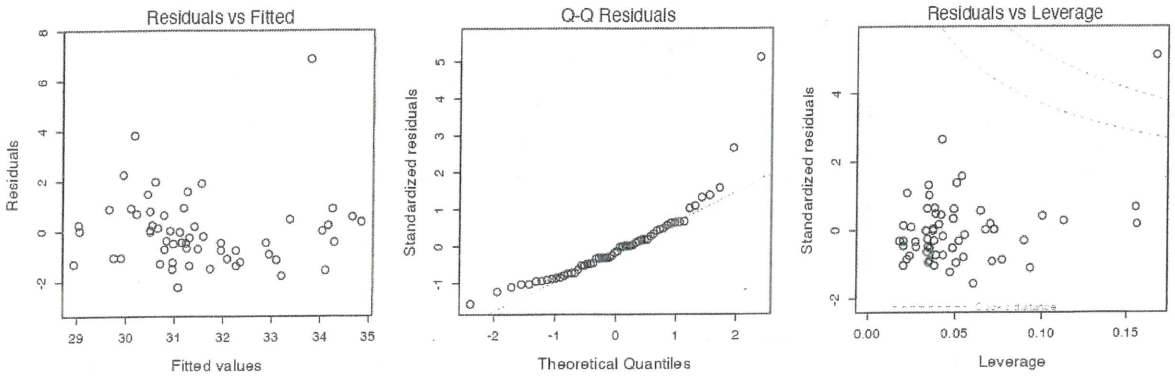
(E) There are no unduly influential points

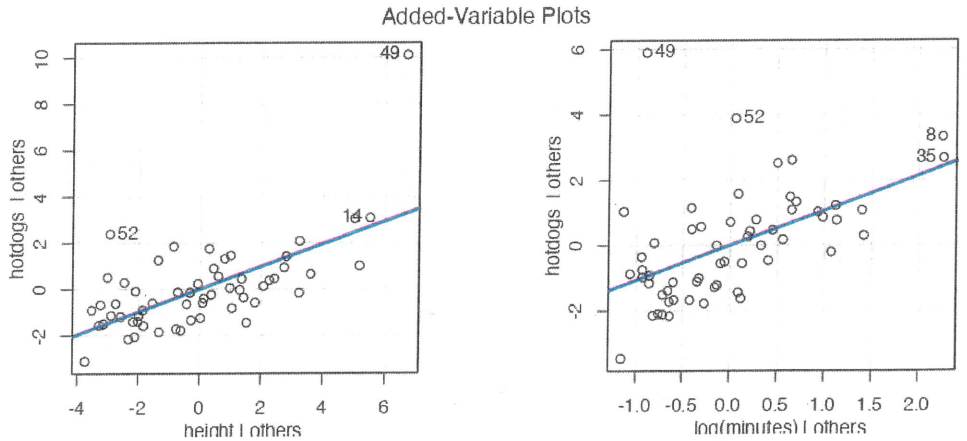
According to the AV Plots, there does seem to be some unduly influential points, esp obs 49.

## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| log(`minutes`) | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

While a BP Test for a model shows equal variance, we still need to look at each models plot to verify equal variance. Because model 1's test is "better" doesn't make it a better model. We need to perform different algorithms and methods as well as understand what we want the model to accomplish before picking a "superior" model.

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Model 2 was transformed which could give it some advantages for model fit and making the assumptions met. The CV RMSE is also smaller, potentially making it better at predicting.

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context

$\hat{\beta}_2 = 1.06702$ Holding height constant, we can expect with a 1% increase that the total amount of hotdogs consumed will increase by approximately $\frac{1.06702}{100}$.

$\approx 0.0106702$

$\hat{Hotdogs} = -6.05589 + 0.45834(68) + 1.06702(\log(79))$

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$-6.05589 + 0.45834(68) + 1.06702(4.369)$

$\hat{Hotdogs} = 31.813$ hotdogs

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error?

$n-p-1$

$58-2-1 = 55$

(A) 55

(B) 57

(C) 56

(D) 48

SE CI SE PI

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1+x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$PI = \hat{y}_0 \pm t^* s\sqrt{1+h_0}$ (SE)

$31.813 \pm 2.004(1.485)$

(28.83706, 34.7889)

# Part III - Application

## Questions 20-24 concern the following dataset:

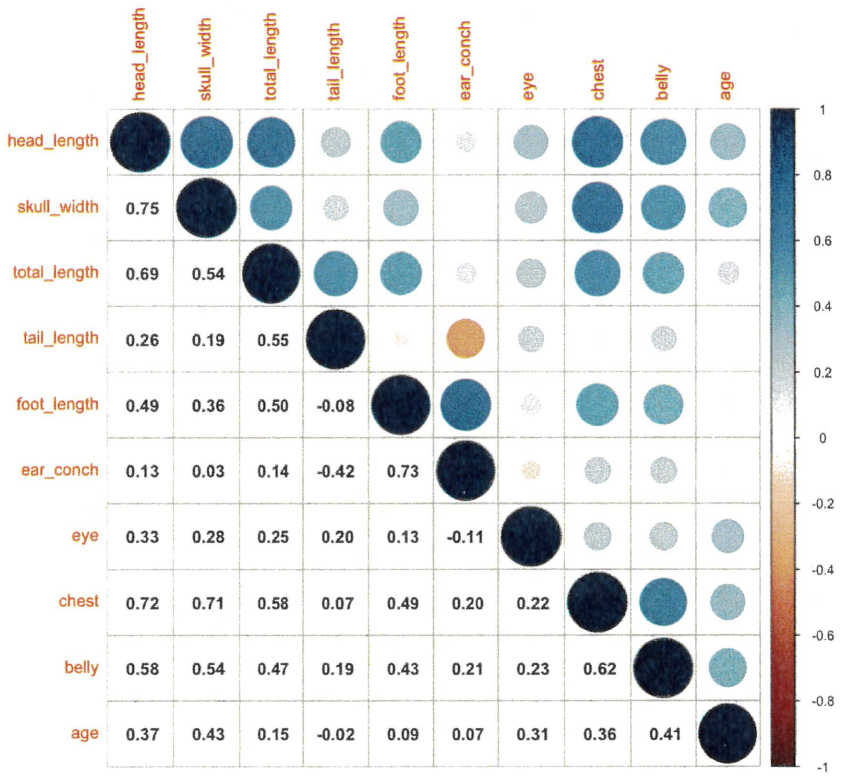
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

| Variable | Description |
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
foot_length   -0.21499    0.10733  -2.003   0.0507 .
ear_conch      0.15730    0.10231   1.538   0.1306
eye            0.43736    0.22549   1.940   0.0582 .
chest          0.01883    0.18154   0.104   0.9178
belly          0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$CI = \hat{\beta}_i \pm t^* SE(\hat{\beta}_i)$

$.16559 \pm 2.01(.10549)$

$[-0.0474499, 0.3786299]$

$.16559 \pm 2.01(.10549)$

$=$

$[-0.0474499, 0.3786299]$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident, holding other variables constant, that the coefficient for belly width is in the interval (-0.023, 0.443).

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

When all the variables are tested together for the overall F test, they show that at least one of the predictors influences the possum's age. Predictors on their own might not by themselves show this, but collectively they can.

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

Best subsets tries every possible combination of predictors. AIC is the metric used to decide which of these models is the best. AIC uses a penalty for more predictors, but not as much as BIC would. AIC is equal to $n\ln(SSE/n) + 2k$. So the model that produces the best AIC is the one that Best Subsets chooses.

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

B Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

# STAT 330-001

40142034 21

## Bagley, Benjamin

282360867 benjibag 801-234-9726

Test: 638191 Form: A
Entry: 04:20pm 10/20/2026
Employee: Brynli P.

Wadsworth, Spencer Gordon
Checked out:

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(B) FALSE

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- TRUE
- FALSE

**Question 3: (Circle one answer)** When adding an additional variable to a linear regression model, the model's $R^2$ ...

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(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate (accurate or precise?) confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

The coefficients are being shrunk towards zero, generally. By adding another term to the regression based on $\lambda$ before minimizing SSE

i.e $\min_{\beta,\beta_0} \left[ \sum_{i=1}^{n} y_i - \beta_0 - x_i^T \beta)^2 + \lambda \sum_j \beta_j^2 \right]$ for ridge regression

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$ $1.44(.333) = .48$

**(Circle one answer)** Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

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(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

**Question 10:**

(A) What is the formula for $\widehat{\beta}$?

$\vec{\beta} = (X^TX)^{-1}X^TY$

(B) What is the conceptual difference between $\widehat{\beta}$ and $\beta$?

$\beta$ is the true coefficient vector of the theoretical model, while $\vec{\beta}$ is the predicted coefficent vector calculated from the data in our sample.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

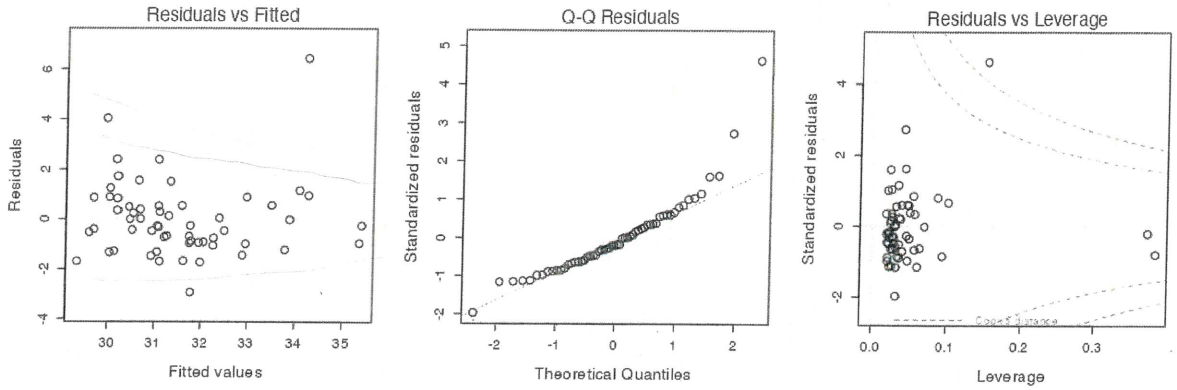
By using a guess of coefficients we get a possible line for our model. We sum the squared of each observation's distance from this line, and minimize that value with calculus. We find the least sum of squares. Conceptually, minimizing the squared residuals means finding the set of coefficients that get as close as possible to each point.

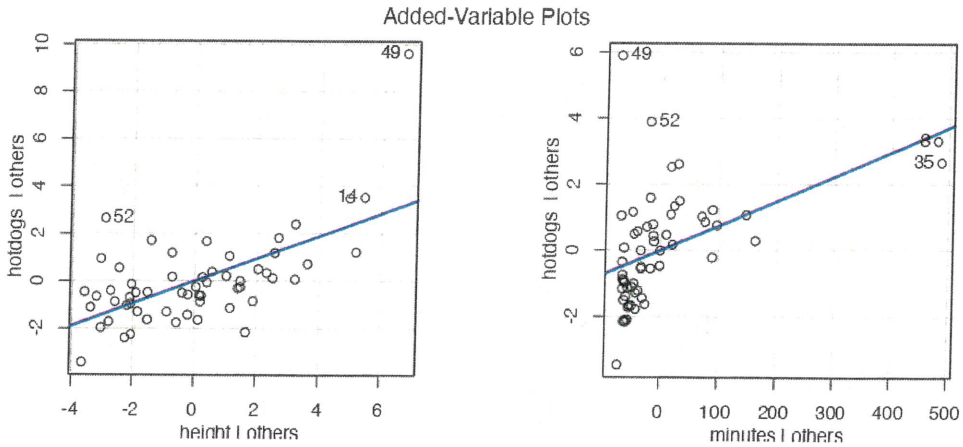
# PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

$\hat{\beta}_1 = .472892$. Holding the minutes spent training constant, every additional inch of height is associated with an increase in the number of hotdogs a contestant can eat of .472892, on average.

**Question 12:** **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0

(B) The 95% confidence interval of $\beta_2$ does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

These data follow a somewhat linear relationship. The residuals v fitted plot shows no alarming signs of curvature. Though the minutes added variable plot doesn't show much of a relationship at all, the AV plot for height shows a decent linear relationship.

(B) There is independence between each observation

The data are not likely to be mostly independent. Competitive eaters, especially those living near each other, probably run in similar circles, maybe they practice together. That means that the minutes spent practicing not independent between contestants.

(C) The errors are normally distributed

The errors do not follow a normal distribution. The QQ plot has an obvious curve, and the Shapiro-Wilk test supports that. With a p-value $< .0001$, we have evidence to reject the null hypothesis that the errors are normally distributed

(D) The errors have constant variance (homoscedasticity)

The errors do not have constant variance. The residuals v. fitted plot shows an obvious funnel shape, especially ignoring the point with a very large residual on the right side. The Breusch-Pagan test does not support this with a p-value of .071, but that is still a low p-value, and the plot does not support the tests conclusion.

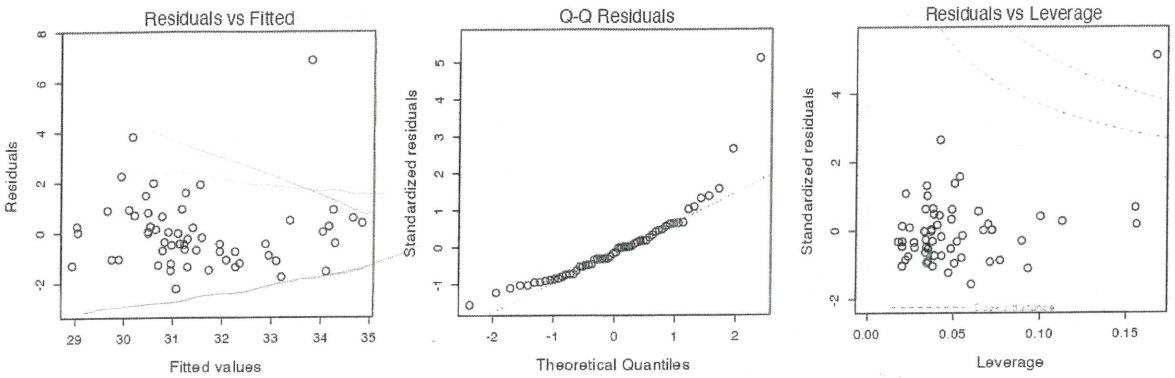
(E) There are no unduly influential points

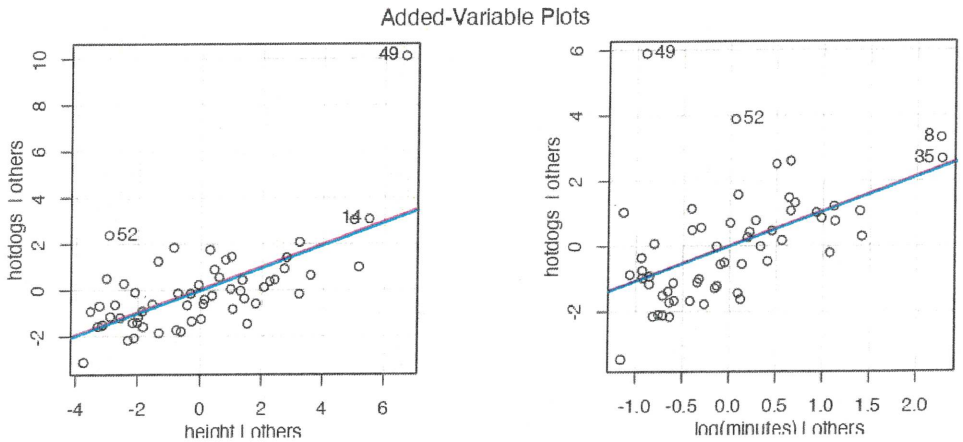
There are a few influential points. The residual v leverage plot shows one worrying point in the top of the plot, and two more on the right side. These points are influential, especially the upper point. The upper point will have the highest cook's distance, and the right points will have high leverage.

## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | Pr($>\|t\|$) |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| log(`minutes`) | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

One p-value is not enough to judge an entire model. Both residuals v. fitted plots show worrying heteroscedasticity, meaning both do not completely meet the equal variance assumption, but both can still be useful.

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Interpretation is always more difficult with a log transform like in model 2. Although the LINE assumptions are marginally better (excepting the B-P test), the improvement is very slight. Most of the 5 plots shown look very similar. Model 2 fulfills LINE assumptions better, but will be more difficult to interpret.

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context

$\beta_2 = 1.06702$. Holding the height of a contestant constant, a one percent increase in the amount of time spent practicing is associated with an increase of 1.06702 hotdogs eaten in 10 minutes, on average.

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$$\text{hotdogs} = -6.05589 + 0.48834(68) + 1.06702\log(79)$$
$$= 31.813 \text{ hotdogs}$$

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error?

$n-p-1 = 58-2-1 = 55$

(A) 55

(B) 57

(C) 56

(D) 48

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1+x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$31.813 \pm 2.004(1.485) \rightarrow (28.837, 34.789)$

# Part III - Application

## Questions 20-24 concern the following dataset:

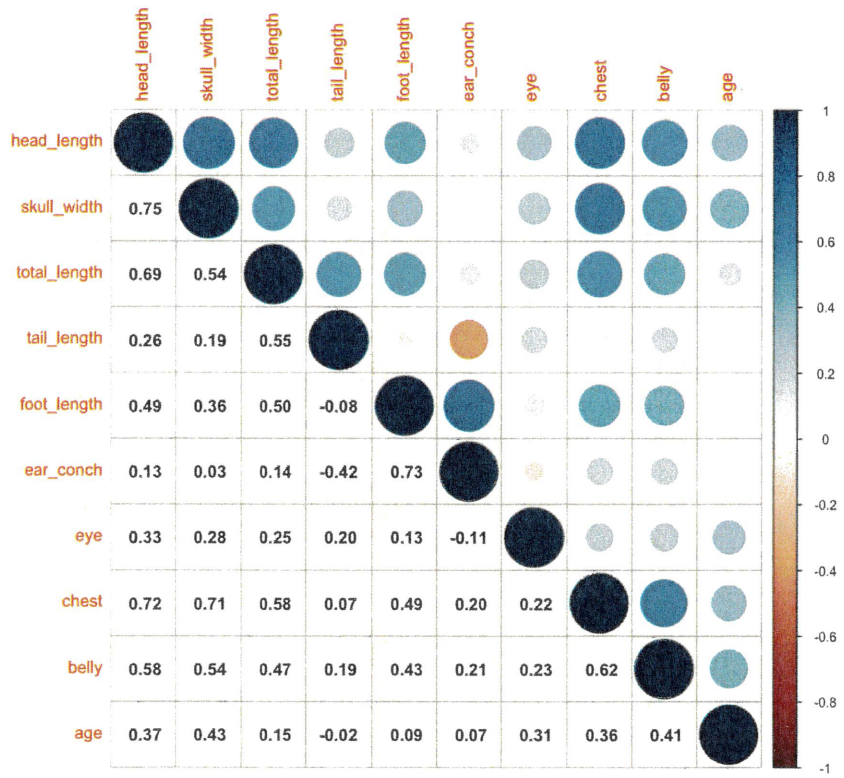
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

| Variable | Description |
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$.16559 \pm 2.01(.10599)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the true $\beta_{belly}$, the true coefficient on belly girth, is between (-0.023, 0.443) cm.

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

Multicollinearity. If variables vary together, they can have a stronger overall effect on the response even if each one individually does not have a significant effect.

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

Make a set of linear models with every possible combination of predictors, and then calculate AIC for each. Keep the model with the lowest AIC.

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

# STAT 330-001

40142034 20

## Lopez Mendoza, Hiram

039550151 hiraml 801-739-0331

Test: 638055 Form: A Wadsworth, Spencer Gordon

Entry: 03:44pm 10/20/2026 Checked out:

Employee: Hannah d.

**SPECIAL INSTRUCTIONS**

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly.
Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

**PLEASE TURN OFF YOUR CELL PHONE**

NS

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020

# STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

**Instructions**

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

## PART I

**Question 1: (Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

**Question 2: (TRUE/FALSE)** If the relationship between $x$ and $y$ is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

**Question 3: (Circle one answer)** When adding an additional variable to a linear regression model, the model's $R^2$ ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

**Question 4: (Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination ($R^2$) to become negative.

**Question 5:** **(Circle all that apply)** Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the $x$'s or $y$ may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of $\beta$ can be obtained by fitting a simple linear regression for each $x_k$ only.

(E) The assumptions on the error variables (i.e., the $\epsilon_i$) in a multiple linear regression model are exactly the same as they are for simple linear regression.

**Question 6:** **(Circle one answer)** What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

**Question 7:** **(Circle all that apply)** In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

LASSO and Ridge regression are often referred to as "shrinkage methods because they shrink the coefficients through a penalty. Ridge and LASSO deliverately introduce bias to reduce variance in a model; they do this by selecting a penalty value ($\lambda$), often being the largest $\lambda$ within 1 std. error from the mean, and as $\lambda$ increases the coefficient, $\beta$, shrinks in a balancing act. Because LASSO penalizes more heavily, some coefficients turn to zero, making LASSO a model selection method. Ridge on the other hand, shrinks the coefficients smoothly, stabilizing the coefficients, but never allowing them to turn to zero.

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$D_i = \frac{2.4^2}{3+1}\left(\frac{.25}{1-.25}\right) \approx 0.48$

$2.4^2 = 5.76$

$4/n = \frac{4}{100} = 0.04$

**(Circle one answer)** Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

**Question 10:**

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = (X^T X)^{-1} X^T Y$

(B) What is the conceptual difference between $\widehat{\beta}$ and $\beta$?

$\beta$ is the coefficient that we want to estimate, and $\hat{\beta}$ is the estimated coefficient of the predictor in our fitted model from the data.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

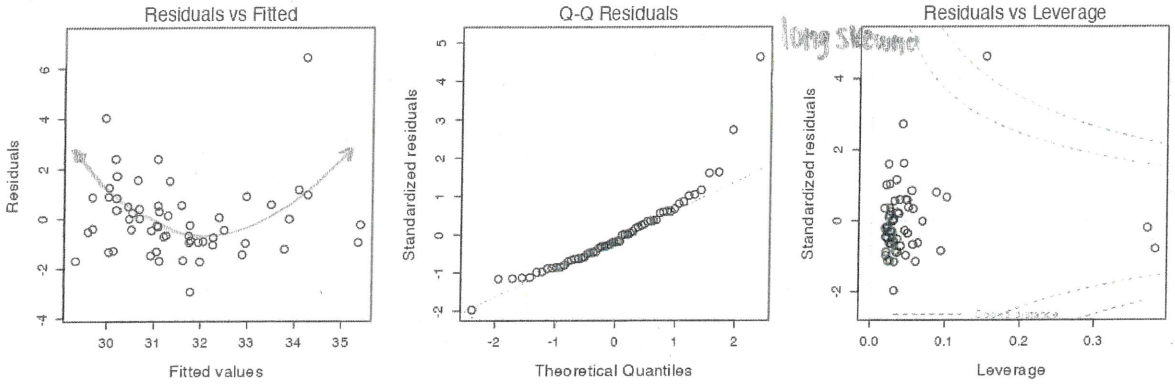
Least squares can be used to estimate regression coefficients by deriving the formula for $\hat{\beta}$ by taking the gradient, setting equal to zero, substituting $\beta$ for $\hat{\beta}$, and solving for $\hat{\beta}$, which allows us to choose the $\hat{\beta}$ with the minimum error.

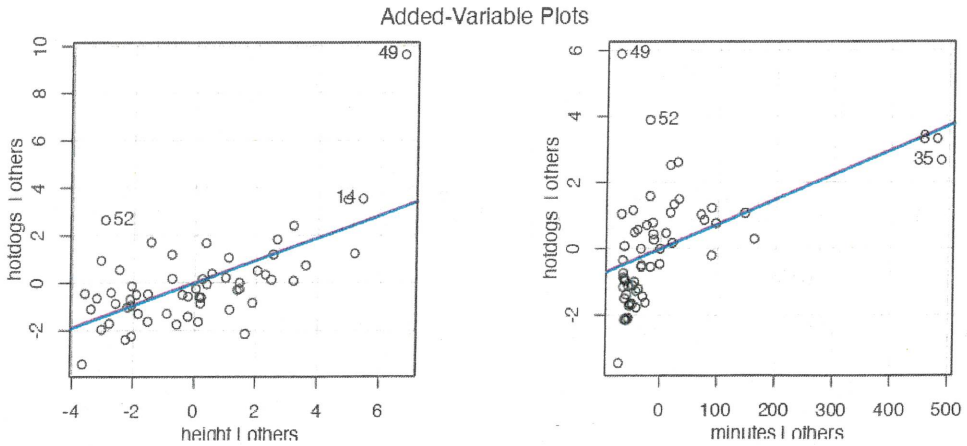
# PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

Holding minutes constant, a one inch increase in height is associated with an average 0.473 hotdog consumption increase.

**Question 12:** **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0

(B) The 95% confidence interval of $\beta_2$ does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$ $\hat{\beta}_1 = 0.4729$ not $\beta_1$

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Analyzing the Residuals vs Fitted Values plot we see the relationship is not linear since the values follow a curved trend that looks like a parabola. The AV plot for minutes also shows that the values do not follow a linear trend.

(B) There is independence between each observation

We cannot assume independence because the observations were not random, and most participants were from 4 specific counties.

(C) The errors are normally distributed

The Q-Q plot shows errors curving away from the line of normality and some are significantly deviated; therefore, the errors are not normally distributed. The Shapiro-Wilk test supports this evidence by rejecting the null hypothesis, concluding that the errors do not come from a normal distribution.

(D) The errors have constant variance (homoscedasticity)

Despite the Breusch-Pagan test saying there is no evidence that the errors are not constant, observations of the variance in AV and Resid vs. Fitted plots suggest that the errors are not constant since the errors change for different fitted values.

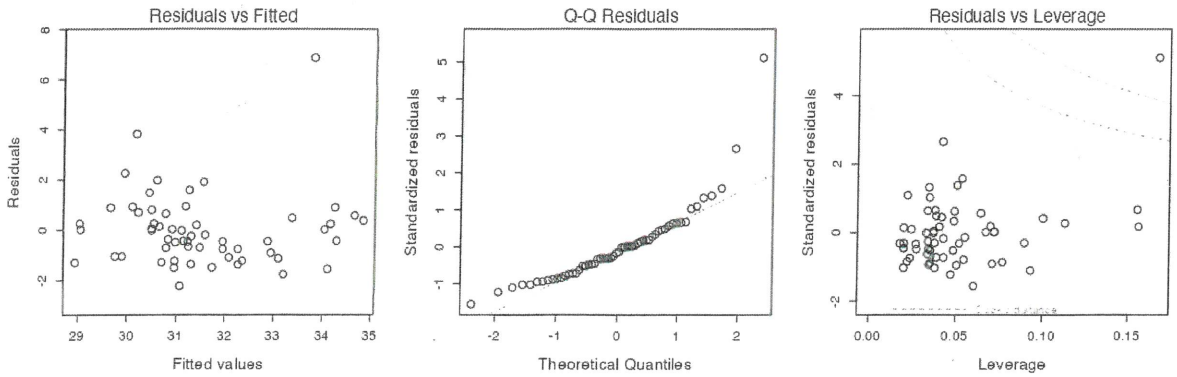
(E) There are no unduly influential points

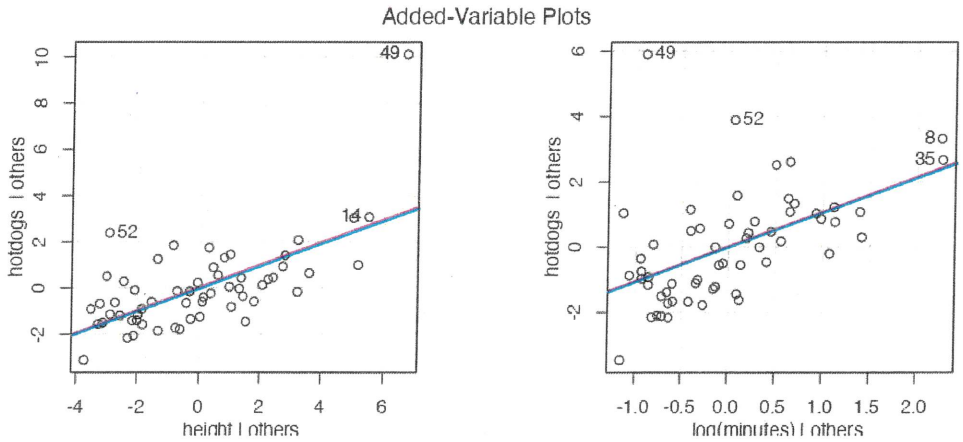
There is one point that seems very influential because it has a very high standardized residual and some leverage. There are also some potentially influential points.

## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| log(`minutes`) | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would respond by saying that is incorrect because relying only on the Breusch-Pagan test does not give reliable information. If we observe the spread of the variance in Resid vs Fitted and AV plots, we find that model 2 has a more equal spread of variance than model 1 despite the BP test telling us the opposite. It is better make assumptions based on our observations than relying on tests.

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Model 2 seems to have more equal variance, and fits the linear model better than model 1. However, errors in Model 2 still deviates from normality about the same as model 1 and still has an influential value.

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context

Level-log $\hat{\beta}_2 = 1.06702$

The estimated value of $\beta_2$ is 1.06702.

Holding height constant, a 10% increase in minute per week training is associated with an estimate $(1.06702)\log(1.10) \approx 0.102$ increase in mean hot-dog consumption.

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that $\log(79) = 4.369$. Use this information and the fit for model (2) to answer the following questions.

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\hat{y} = (-6.05589) + 0.48834(68) + 1.06702 \log(79)$

$\hat{y} \approx 31.81$

The predicted number of hotdogs my friend can consume in 10 minutes is about 31.81.

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

$n-p-1$

$58-2-1=55$

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

95% Prediction C.I. $= \hat{y} \pm SE \cdot t$

$= 31.81 \pm 1.485(2.004)$

95% Prediction C.I. : (28.83406, 34.78594)

# Part III - Application

## Questions 20-24 concern the following dataset:

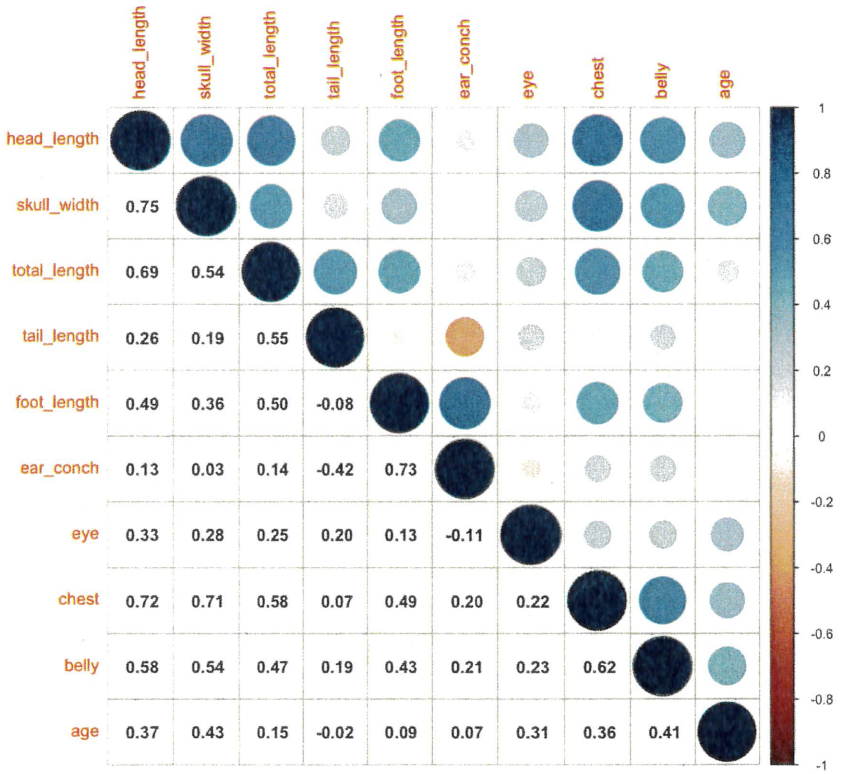
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

| Variable | Description |
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

95% C.I.:

$0.16559 \pm 0.10559 \cdot (2.01)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the true mean belly width value falls between -0.023 and 0.443 cm.

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

None of the individual coefficients have a significant effect at the $\alpha = 0.05$ level because all together they only explain a small portion of the total variance. $R^2$ measures that all the predictors together only explain about 35% of the variance in a possum's age; therefore, their coefficients are not found to be significant.

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

The best subsets algorithm goes through every single possible model comparing predictors and models by comparing their own AIC score, the model with the lowest AIC "score" is then selected as the most optimal model.

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

D Choose the model with the largest RMSE

A Choose the model with the smallest RMSE

A Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

A Compute each model's AIC and BIC

B Fit each model using the testing data

B Fit each model using the training data

D Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

# STAT 330-001

40142034 19

## Yoshizawa, Towa

737254279 ty243 385-254-1935

Test: 638040 Form: A Wadsworth, Spencer Gordon

Entry: 03:41pm 10/20/2026 Checked out:

Employee: Maxie E.

**SPECIAL INSTRUCTIONS**

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

**PLEASE TURN OFF YOUR CELL PHONE**

NS

40142034

019

# STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

**Instructions**

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

## PART I

**Question 1:** **(Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

**Question 2:** **(TRUE/FALSE)** If the relationship between $x$ and $y$ is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

**Question 3:** **(Circle one answer)** When adding an additional variable to a linear regression model, the model's $R^2$ ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

**Question 4:** **(Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination ($R^2$) to become negative.

**Question 5:** **(Circle all that apply)** Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the $x$'s or $y$ may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of $\beta$ can be obtained by fitting a simple linear regression for each $x_k$ only.

(E) The assumptions on the error variables (i.e., the $\epsilon_i$) in a multiple linear regression model are exactly the same as they are for simple linear regression.

**Question 6:** **(Circle one answer)** What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

**Question 7:** **(Circle all that apply)** In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS bias ↑ ?

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

They shrink the regression coefficient toward 0.

How? → By minimizing the sum of squard error plus a penalty on the size of the coefficients

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

**(Circle one answer)** Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

**Question 10:**

(A) What is the formula for $\widehat{\beta}$?

$D_i = \frac{(2.4)^2}{3+1}\left(\frac{0.25}{1-0.25}\right)$

$= \frac{5.76}{4} = \frac{1.44\ (0.3333)}{} = 0.4795\ldots$

(B) What is the conceptual difference between $\widehat{\beta}$ and $\beta$?

$\widehat{\beta}$ has a "^" which means estimation.

$\beta$ is used theoretical linear model. They have "$\epsilon$".

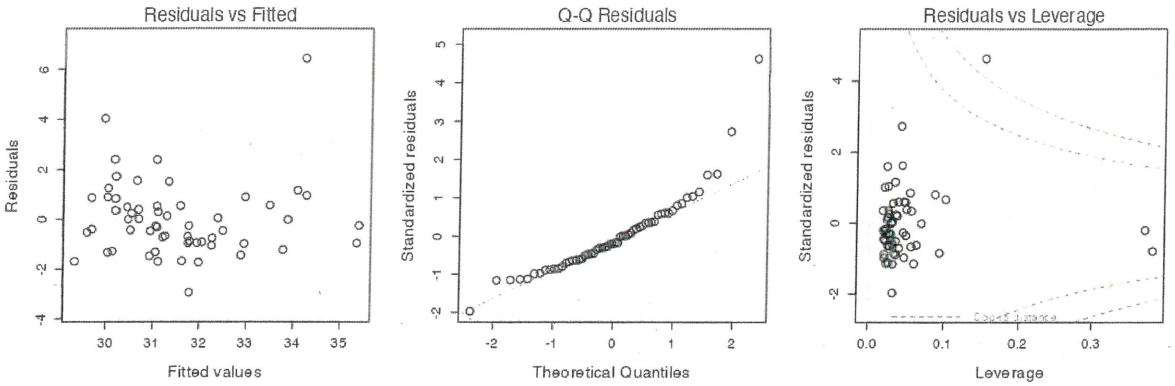
(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

# PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

$\beta_1 = 0.472892$

For every one hotdog consumed in a single (0 minute, the average (expected) contestant's height in inches increase by 0.472892.

**Question 12:** **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0

(B) The 95% confidence interval of $\beta_2$ does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Linearity assumption is not violated.

(B) There is independence between each observation

Since Data were collected on 58 different amateur competitive eaters, Independence isn't violated.

(C) The errors are normally distributed

As Theoretical Quantiles increas, residuals rapidly increase (Q-Q Residuals) ——> Normality is likely violated

(D) The errors have constant variance (homoscedasticity)

(Residuals vs Fitted)

The graph (data) has not homoscedasticity —> Equal Variance is likely violated.

(E) There are no unduly influential points

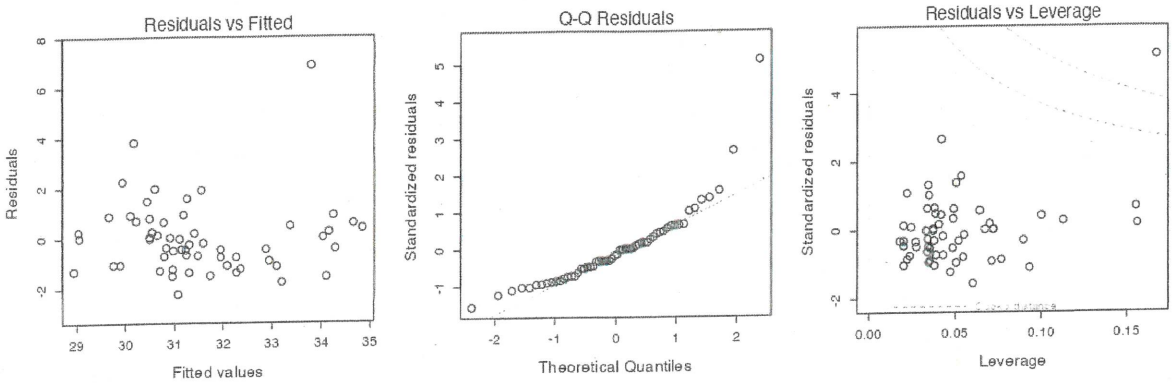
look at "Residals vs Leverage" data, some points are too high.

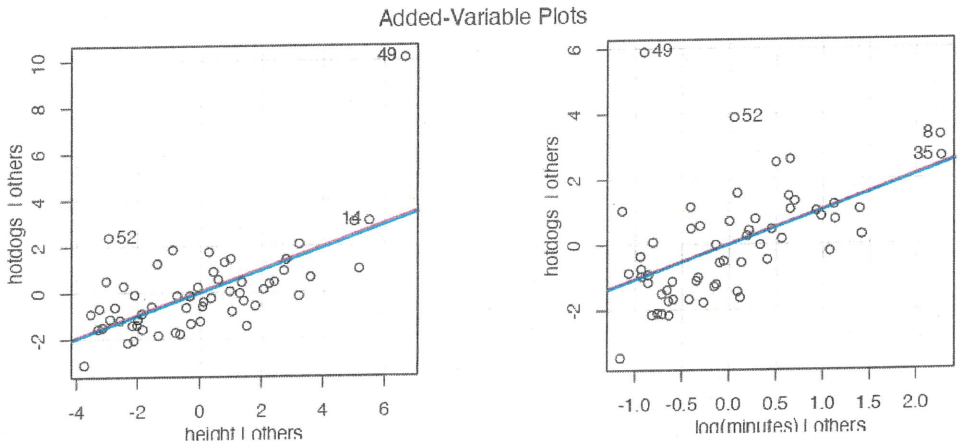
The assumption is likely violated.

## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| log(`minutes`) | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

Compared to model (1), model (2) is easy to interpret.

⟶ Data (Residuals vs Leverage) and (Added-Variable plots)

Model (1)'s data focuses on left side of the graph

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Advantage: easy to interpret.

disadvantage: lower Breusch-pegan p-value than model (1).
higher Std. Error

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context

$\beta_2 = 1.0670^2$

Holding all else constant, a 1 log(minutes) spending training time associated with the total number of hot dogs consumed

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

hot dogs = $\beta_0 + \beta_1 \text{height} + \beta_2 \log(\text{minutes})$

$= -6.05389 + 68 + 4.369 = 66.31311$

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$66.31311 \pm 2.004 \times 0.204 \times 1.485$

$\pm 0.607$

$(65.70611 \;,\; 66.92011)$

# Part III - Application

## Questions 20-24 concern the following dataset:

In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

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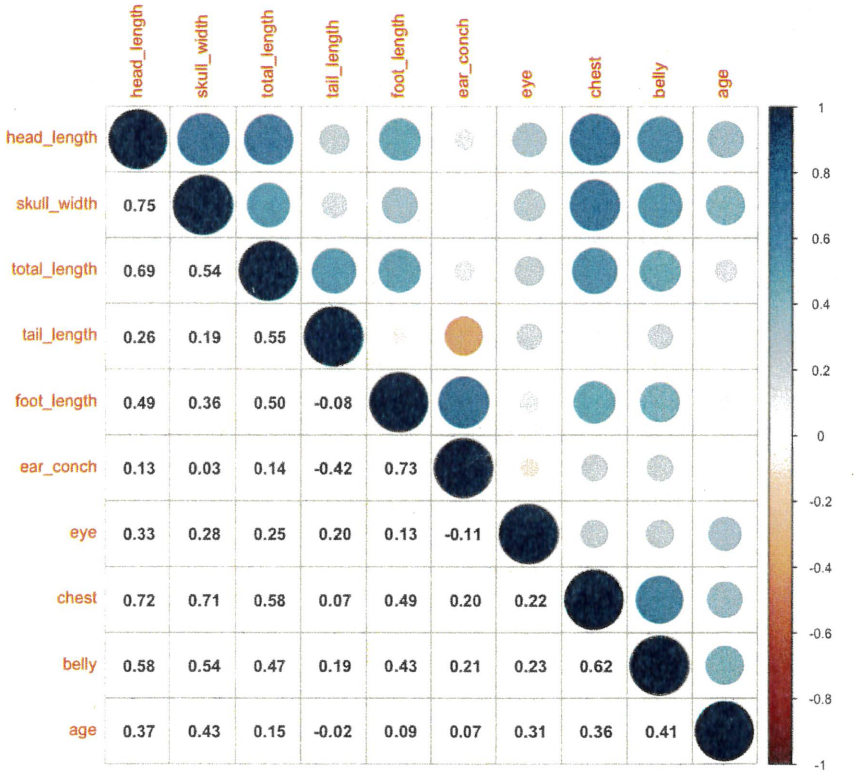
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

Ext

skull_width 0.16559

0.16559 ± 2.01 × 0.10599

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the belly width with an animals age will fall between -0.023 and 0.443.

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

Multicollinearity.

F-test asks whether the predictors are jointly useful, which they are.

→ T-test is different meaning.

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

It fits every possible model to fall data.

→ select the lowest AIC model.

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

A Average RMSE across all test-training splits and report

B Average the predictions and record their value

B Choose the model with the largest RMSE

B Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

B Compute each model's AIC and BIC

C Fit each model using the testing data

C Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

D Split the data into test and training sets

D Use the models fitted to the training data to get predictions for the test data

END OF EXAM

# STAT 330-001

40142034 18

## McCauley, Mitchell

229926433 mmccaul2 916-765-8281

Test: 638007 Form: A Wadsworth, Spencer Gordon

Entry: 03:34pm 10/20/2026 Checked out:

Employee: Justin E.

**SPECIAL INSTRUCTIONS**

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly.
Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

**PLEASE TURN OFF YOUR CELL PHONE**

40142034

NS

018

# STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

**Instructions**

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

## PART I

**Question 1: (Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

**Question 2: (TRUE/FALSE)** If the relationship between $x$ and $y$ is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

**Question 3: (Circle one answer)** When adding an additional variable to a linear regression model, the model's $R^2$ ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

**Question 4: (Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination ($R^2$) to become negative.

**Question 5: (Circle all that apply)** Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the $x$'s or $y$ may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of $\beta$ can be obtained by fitting a simple linear regression for each $x_k$ only.

(E) The assumptions on the error variables (i.e., the $\epsilon_i$) in a multiple linear regression model are exactly the same as they are for simple linear regression.

**Question 6: (Circle one answer)** What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

**Question 7: (Circle all that apply)** In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

They "shrink" the coefficients in the model. Ridge and LASSO reduce the coefficients smoothly toward 0 and LASSO can even make some equal to 0. They do this by allowing increased bias in order to improve prediction.

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

1.44 $\quad 1.44\left(\frac{0.25}{0.75}\right) = 0.48 \quad \frac{4}{100} = .04$

**(Circle one answer)** Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

**Question 10:**

(A) What is the formula for $\widehat{\beta}$?

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

(B) What is the conceptual difference between $\widehat{\beta}$ and $\beta$?

$\hat{\beta}$ is the fitted coefficient estimates based on the data while $\beta$ is the population coefficients.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

Least squares is trying to minimize the error between a data point ($Y_i$) and it's estimated point from the model ($\hat{Y}_i$). Using the sum of the least squares, we choose coefficients that make the distance between the two points as small as possible.

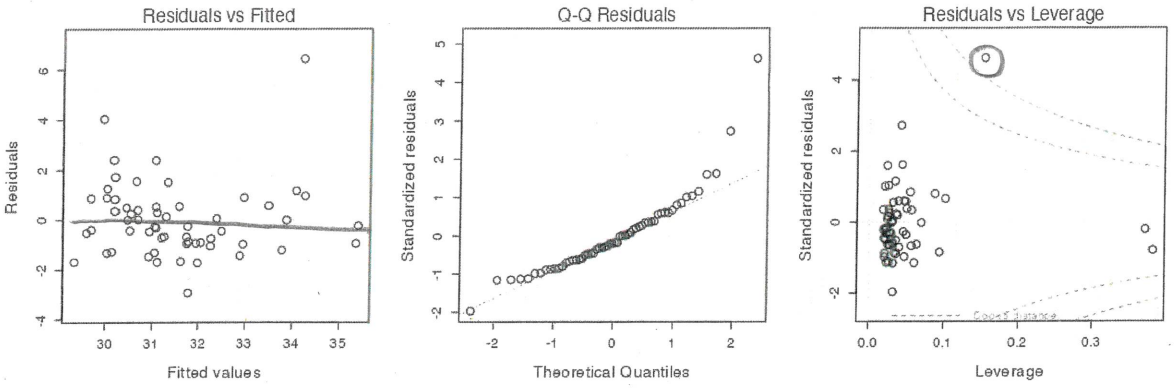
# PART II

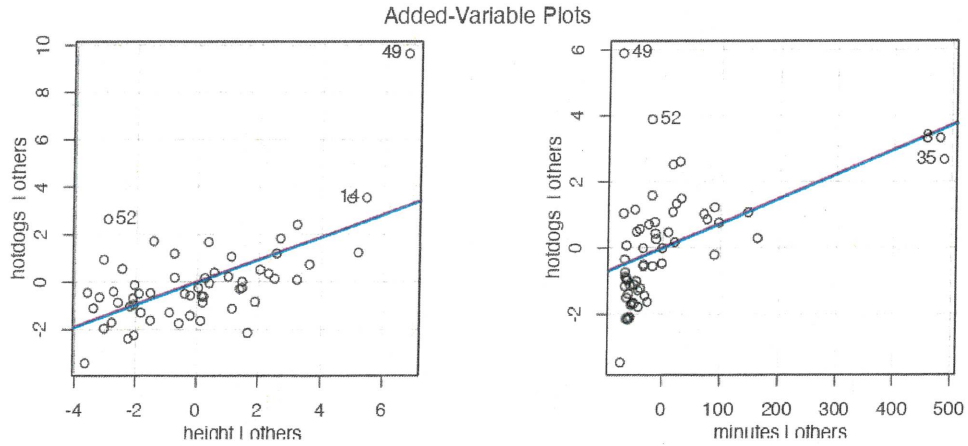
Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

n = 58

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

$\hat{\beta}_1 = 0.47$

Holding minutes trained constant, as we increase a contestant's height by 1 inch, the amount of hotdogs they can eat increase by 0.47 on average.

**Question 12:** **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0

(B) The 95% confidence interval of $\beta_2$ does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

$\hat{\beta}_2 = 0.007417$ $.007417 \pm .00741081$

$SE = .00187$

$t = 3.963$

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Overall, you could say the relationships are somewhat linear. The Residuals vs. fitted plot shows generally a patternless, horizontal cloud of points centered around 0 and the AV plots generally follow a straight line. Minutes trained others less so as there's a gap between the extremes.

(B) There is independence between each observation

Thinking critically, I would say that each competitor is independent from the others because they are different people. They are from similar counties, but I would assume independance

(C) The errors are normally distributed

According to the Shapiro-Wilk test, the low p-value tells us this distribution is nonnormal. But looking at the Q-Q plot, they generally follow the 45° line with some variation on the tails which implies there is skewness. I would hold judgment and look at a histogram

(D) The errors have constant variance (homoscedasticity)

The Breusch Pagan p-value is .071 which tells me the errors have somewhat constant variance. Furthermore, the residuals vs fitted plot shows a pretty equal spread along the horizontal line

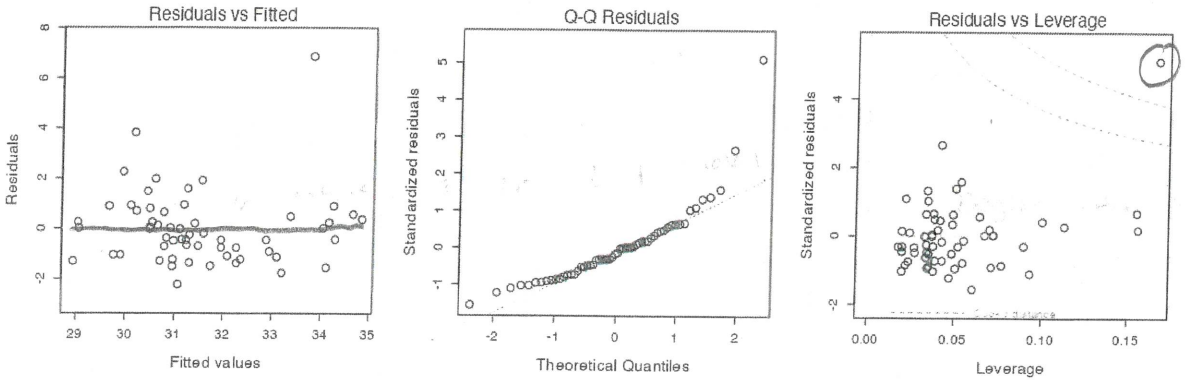
(E) There are no unduly influential points

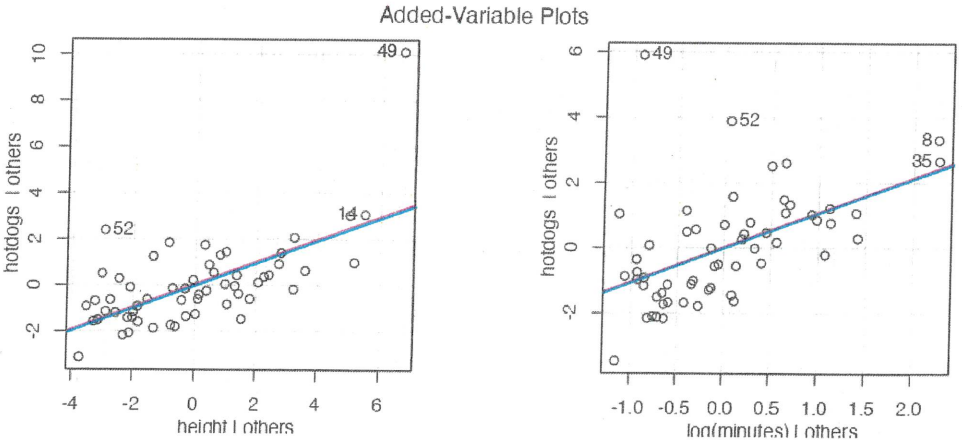
There might be an influential point as there's one in the upper right of the residuals vs leverage plot, but I would investigate that data point to make sure it's entered correctly before I remove it. I do notice it's above our Cook's distance rule of thumb threshold.

## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| log(`minutes`) | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would say that we can't jump to the conclusion just based off the p-value of the Breusch-Pagan test. While Model 2 might have unequal variance between the residuals, it might impact model 1 in other ways. We should also look at the fitted v residuals plot to check the spread.

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Advantages:

- the linearity of the relationship between minutes/others is definitely improved in Model 2
- the CVRMSE value decreased in model 2 which shows that it might be better for predicting

Disadvantages

- equal variance assumption might be violated in Model 2 according to BP test, one point has increased leverage in Model 2 and might be more influential

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context

$\hat{\beta}_2 = 1.07$

Holding height constant, a 1% increase in training time can lead to an average increase of 0.005 hot dogs consumed.

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

hotdogs $= -6.05589 + 0.48834(68) + 1.06702\log(79)$

$= -6.05589 + 33.20712 + 4.6618$

$= 31.81 \approx$ 32 hot dogs

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

$n - p - 1$

$58 - 2 - 1$

$55$

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$t^* = 2.004$

estimate $= 32$

$32 - 2.004(0.204) = 29.5$

$32 + 2.004(e^{1.485})$

(29.5, 40.85)

# Part III - Application

## Questions 20-24 concern the following dataset:

In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

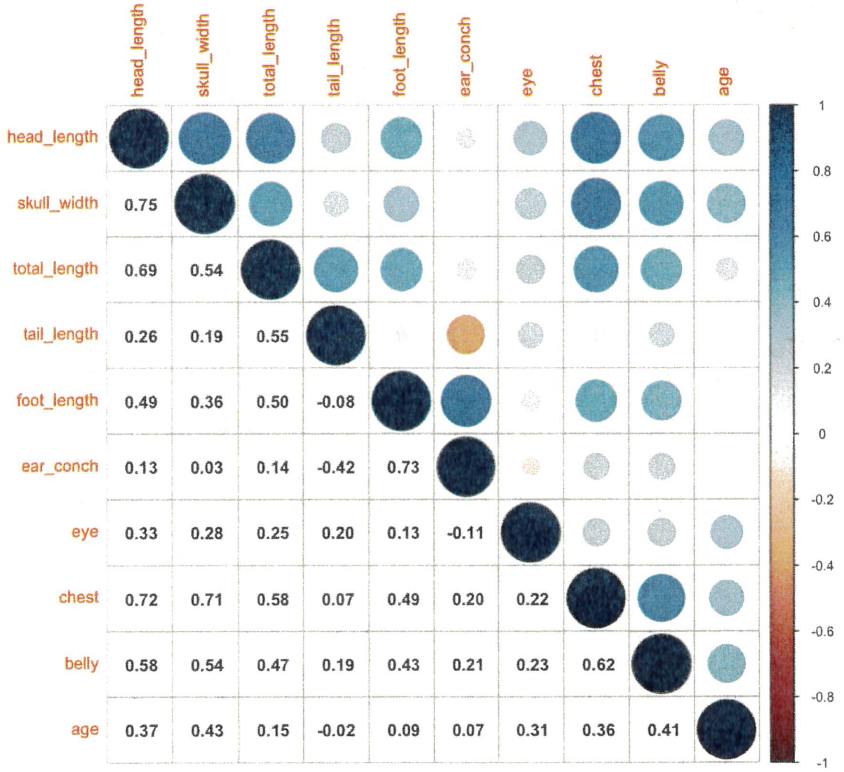
n = 59

| Variable | Description |
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$0.166 \pm 2.01(0.106)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that holding all other predictors constant, a 1 cm increase in belly girth leads to an average change of age between -0.023 and 0.443,

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

Some of the predictors are correlated with one another which would have an effect on the individual model coefficients.

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

The best subsets algorithm goes through all of the possible models and selects the best few that have the lowest AIC values which means that these models are preferable for prediction.

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
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eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

B Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

B Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM